Simple Stereo

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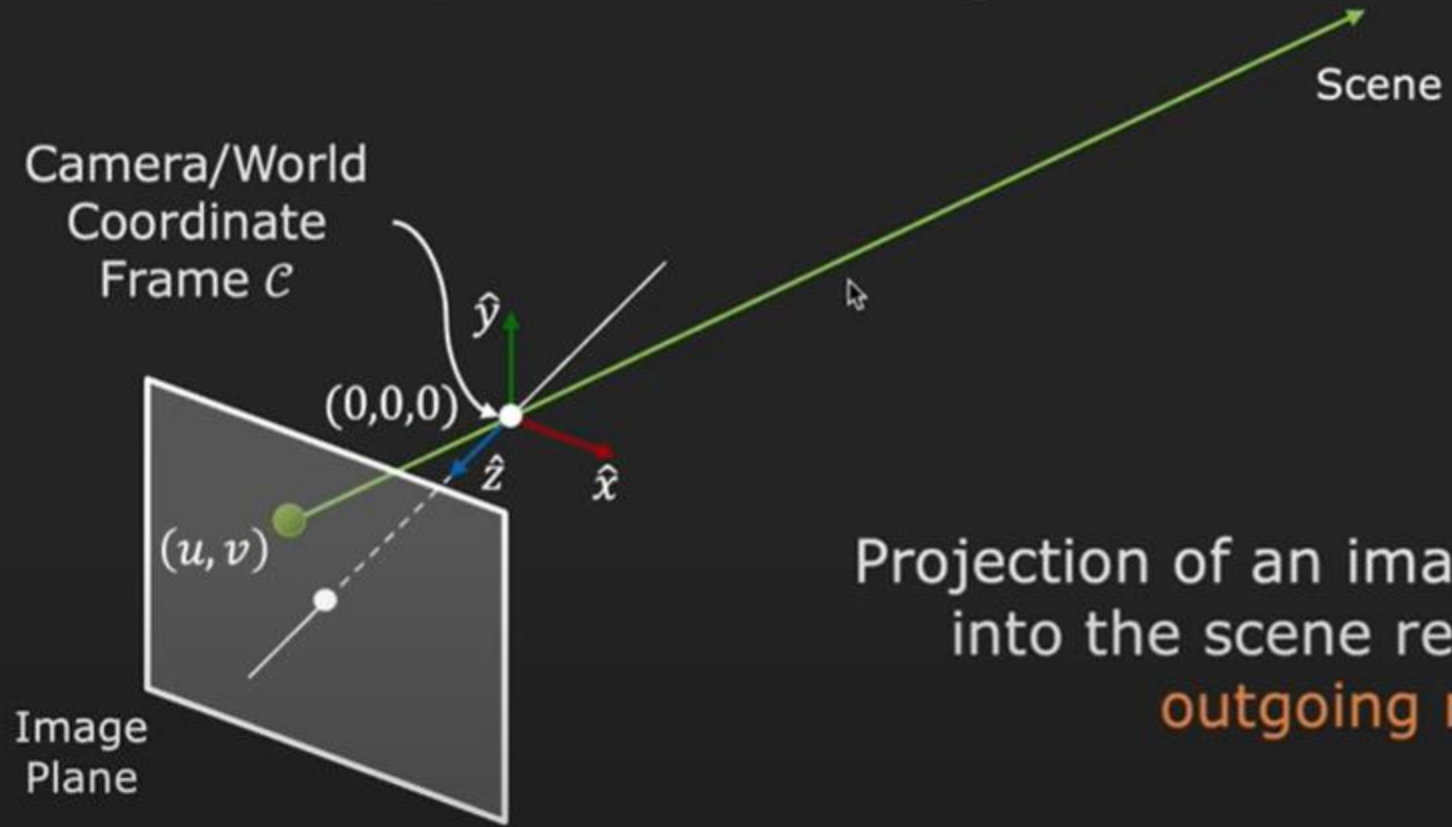
Columbia University

Topic: Camera Calibration, Module: Reconstruction II

First Principles of Computer Vision

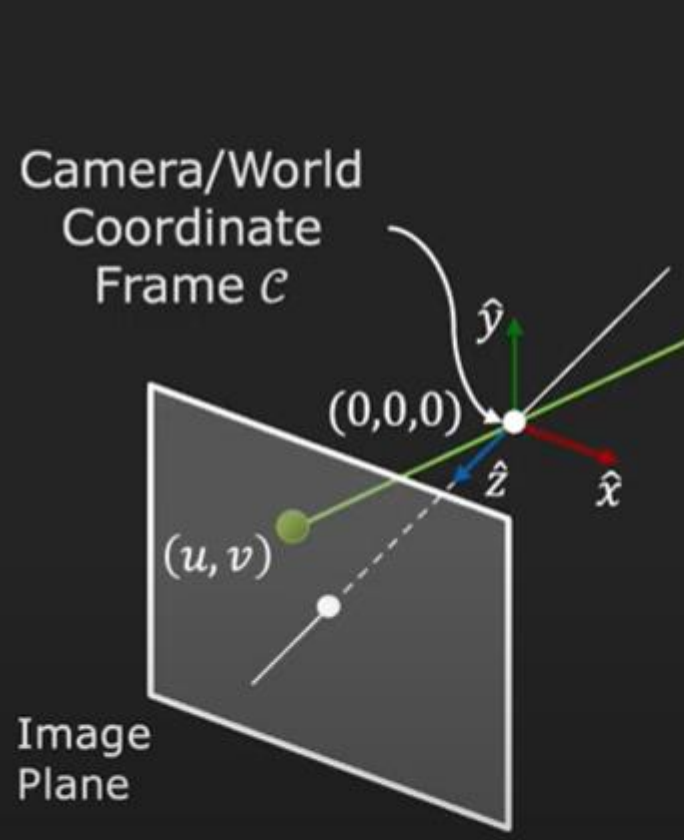
Backward Projection: From 2D to 3D

Given a calibrated camera, can we find the 3D scene point from a single 2D image?



Projection of an image point back into the scene results in an outgoing ray.

Computing 2D-to-3D Outgoing Ray



3D-to-2D:
(Point)

$$u = f_x \frac{x_c}{z_c} + o_x$$

$$v = f_y \frac{y_c}{z_c} + o_y$$

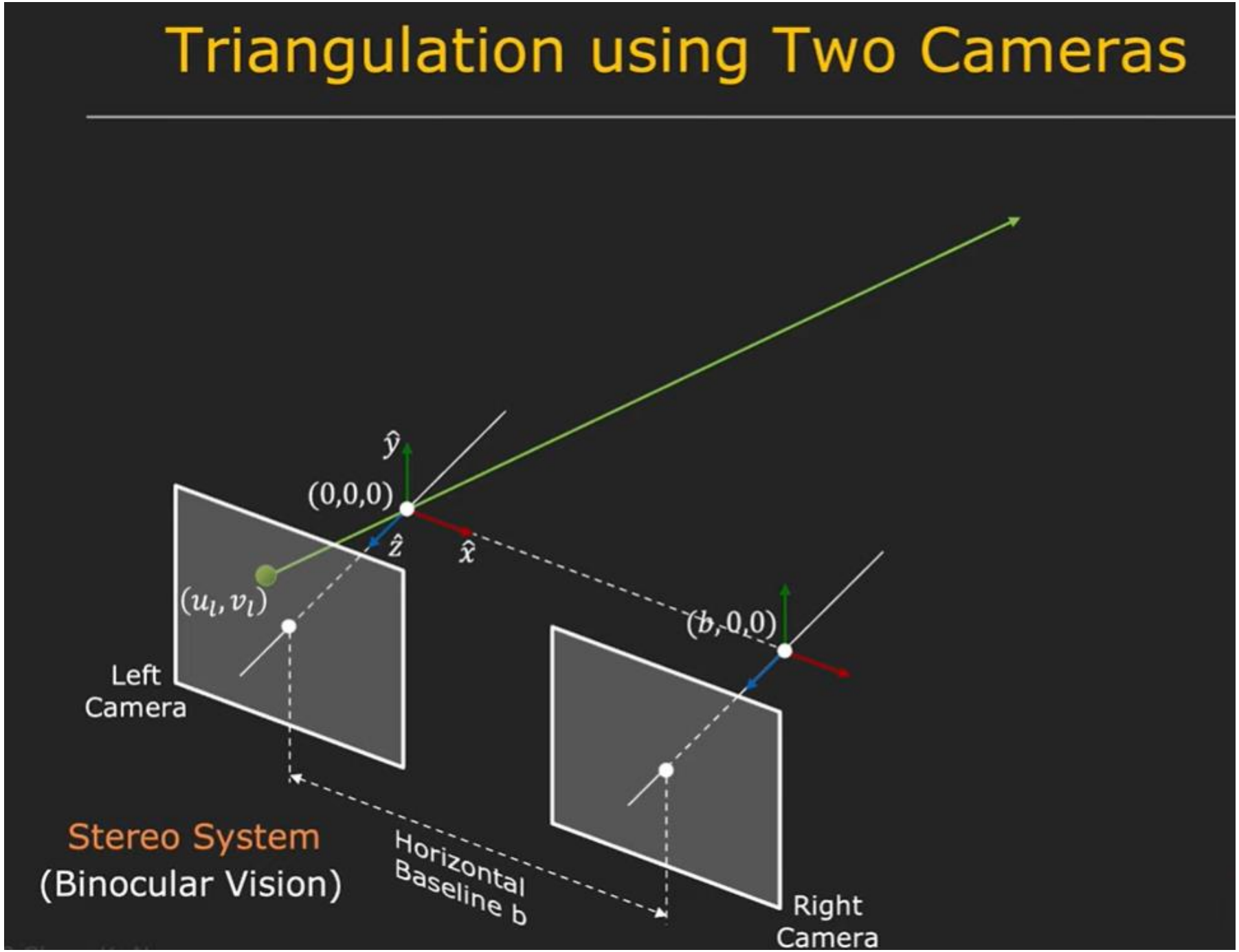
2D-to-3D:
(Ray)

$$x = {}^{z}/_{f_x} (u - o_x)$$

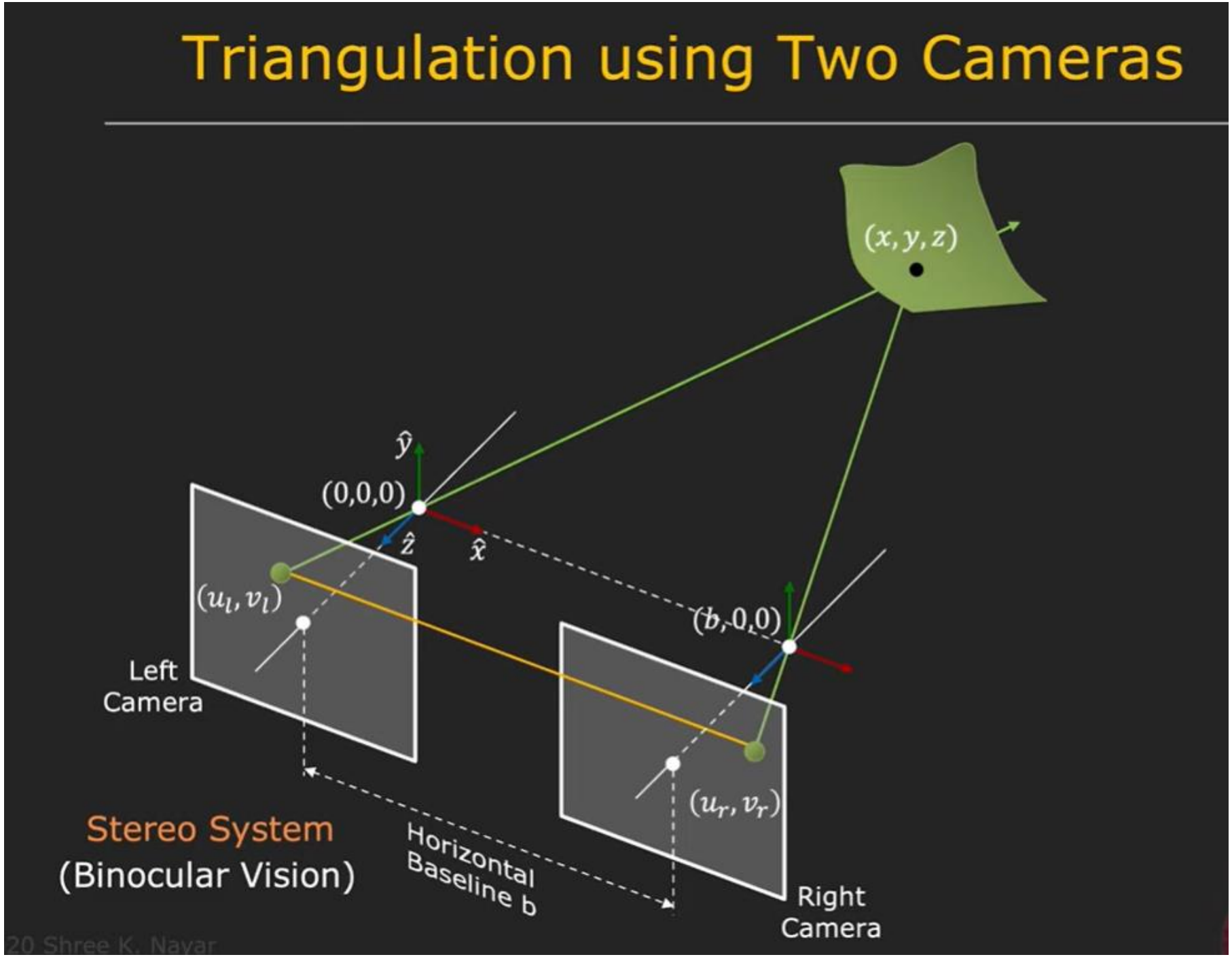
$$y = {}^{z}/_{f_y} (v - o_y)$$

$$z > 0$$

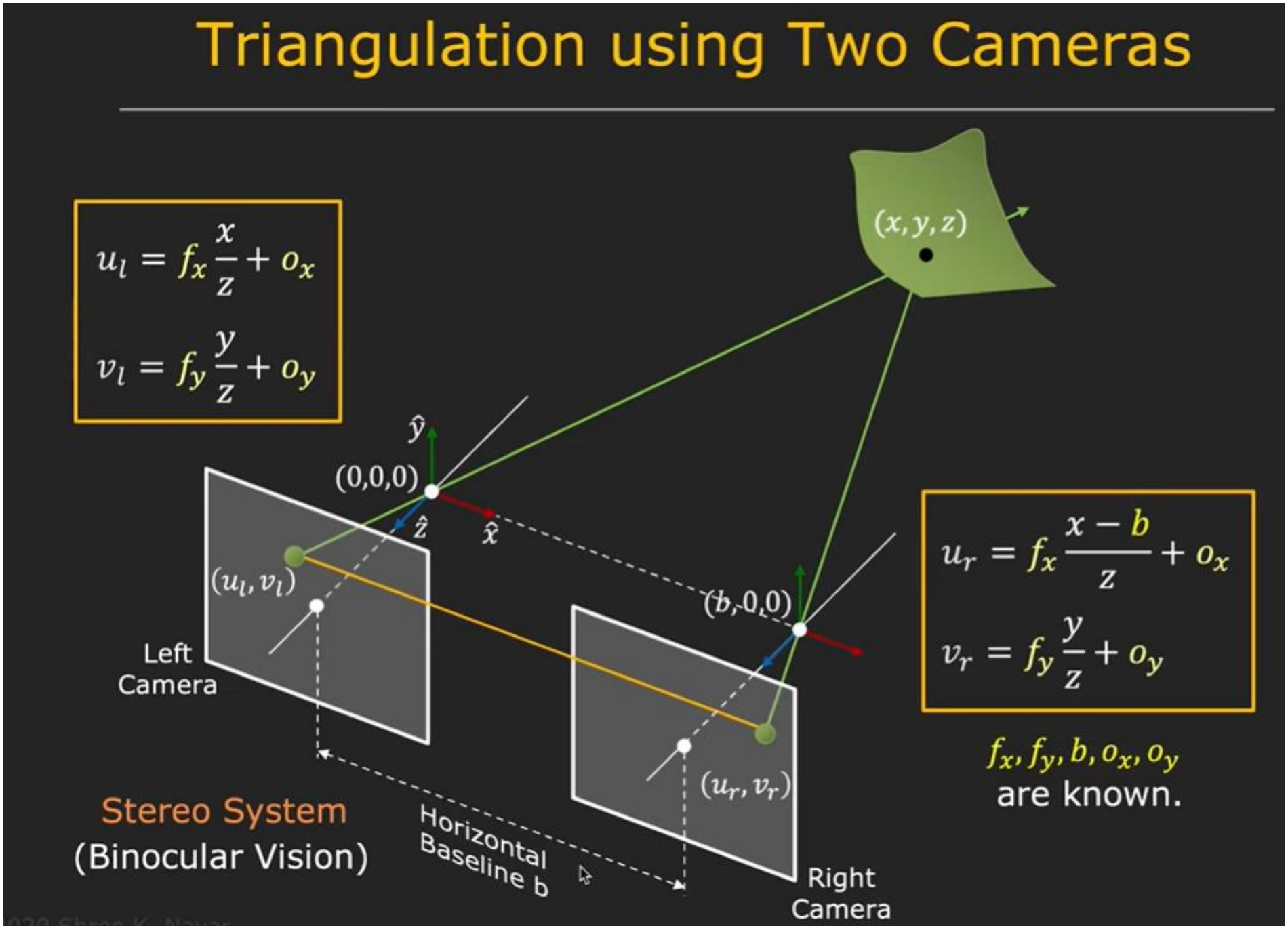
Triangulation using Two Cameras



Triangulation using Two Cameras



Triangulation using Two Cameras



Simple Stereo: Depth and Disparity

From perspective projection:

$$(u_l, v_l) = \left(f_x \frac{x}{z} + o_x, f_y \frac{y}{z} + o_y\right) \qquad (u_r, v_r) = \left(f_x \frac{x-b}{z} + o_x, f_y \frac{y}{z} + o_y\right)$$

Solving for (x, y, z):

$$x = \frac{b(u_l - o_x)}{(u_l - u_r)} \qquad y = \frac{b f_x (v_l - o_y)}{f_y (u_l - u_r)}$$

$$\boxed{z = \frac{b f_x}{(u_l - u_r)}}$$

Simple Stereo: Depth and Disparity

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Solving for (x, y, z):

$$x = \frac{b(u_l - o_x)}{(u_l - u_r)} \qquad y = \frac{b f_x (v_l - o_y)}{f_y (u_l - u_r)} \qquad \boxed{z = \frac{b f_x}{(u_l - u_r)}}$$

where $(u_l - u_r)$ is called Disparity.

Depth z is inversely proportional to Disparity.

Simple Stereo: Depth and Disparity

From perspective projection:

$$(u_l, v_l) = \left(f_x \frac{x}{z} + o_x, f_y \frac{y}{z} + o_y\right) \qquad (u_r, v_r) = \left(f_x \frac{x-b}{z} + o_x, f_y \frac{y}{z} + o_y\right)$$

Solving for (x, y, z):

$$x = \frac{b(u_l - o_x)}{(u_l - u_r)} \qquad y = \frac{b f_x (v_l - o_y)}{f_y (u_l - u_r)} \qquad \boxed{z = \frac{b f_x}{(u_l - u_r)}}$$

where $(u_l - u_r)$ is called Disparity.

Depth z is inversely proportional to Disparity.

Disparity is proportional to Baseline.

Goal: Find the disparity between left and right stereo pairs.



Left/Right Camera Images



Disparity Map (Ground Truth)

Stereo Matching: Finding Disparities

Goal: Find the disparity between left and right stereo pairs.



Left/Right Camera Images



Disparity Map (Ground Truth)

From perspective projection: $v_l = v_r = f_y \frac{y}{z} + o_y$

Corresponding scene points lie on the same horizontal scan line.

Window Based Methods

Determine Disparity using Template Matching

Template Window T



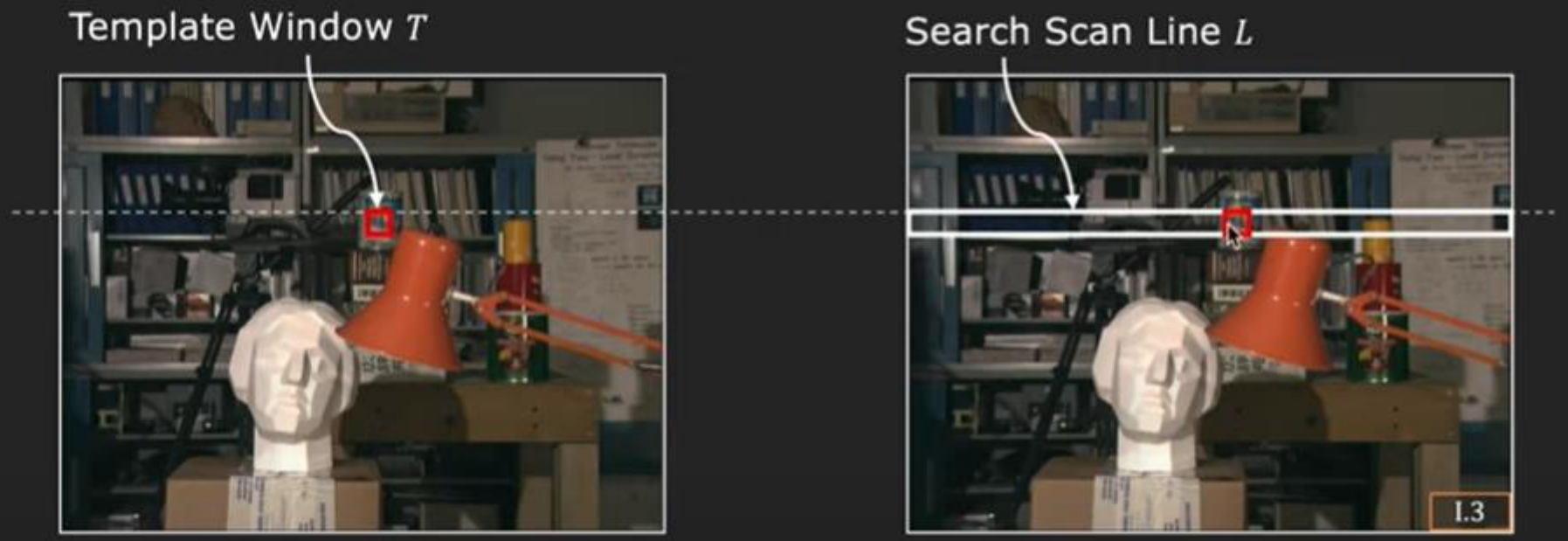
Left Camera Image E_l



Right Camera Image E_r

Window Based Methods

Determine Disparity using Template Matching

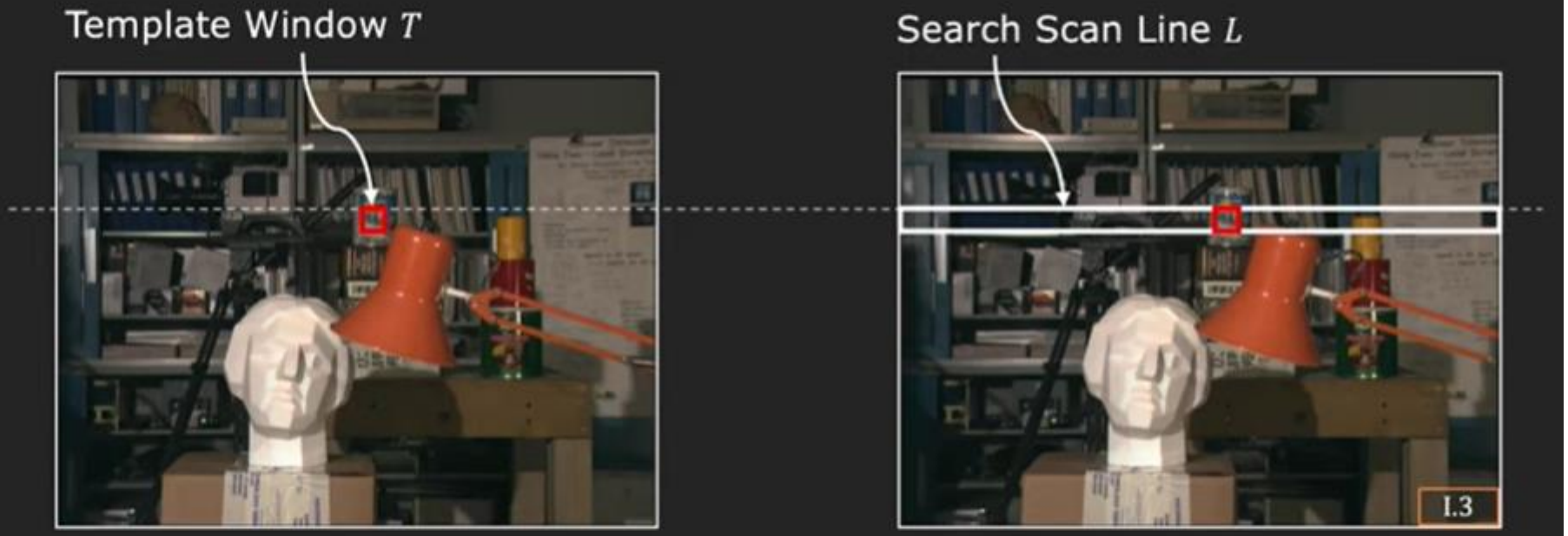


Left Camera Image E_l

Right Camera Image E_r

Window Based Methods

Determine Disparity using Template Matching



Left Camera Image E_l

Right Camera Image E_r

Disparity: $d = u_l - u_r$

Depth: $z = \frac{bf_x}{(u_l - u_r)}$

Similarity Metrics for Template Matching

Find pixel $(k,l) \in L$ with Minimum Sum of Absolute Differences:

$$SAD(k,l) = \sum_{(i,j)\in T} |E_l(i,j) - E_r(i+k, j+l)|$$

Find pixel $(k,l) \in L$ with Minimum Sum of Squared Differences:

$$SSD(k,l) = \sum_{(i,j)\in T} |E_l(i,j) - E_r(i+k, j+l)|^2$$

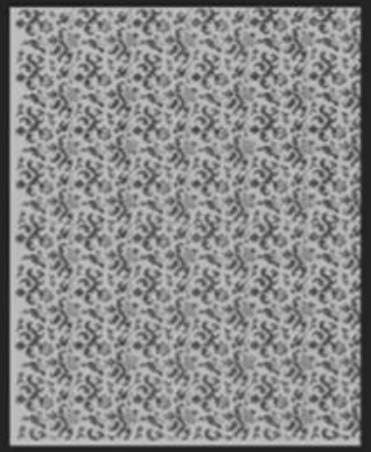
Find pixel $(k,l) \in L$ with Maximum Normalized Cross-Correlation:

$$NCC(k,l) = \frac{\sum_{(i,j)\in T} E_l(i,j)\, E_r(i+k, j+l)}{\sqrt{\sum_{(i,j)\in T} E_l(i,j)^2 \sum_{(i,j)\in T} E_r(i+k, j+l)^2}}$$

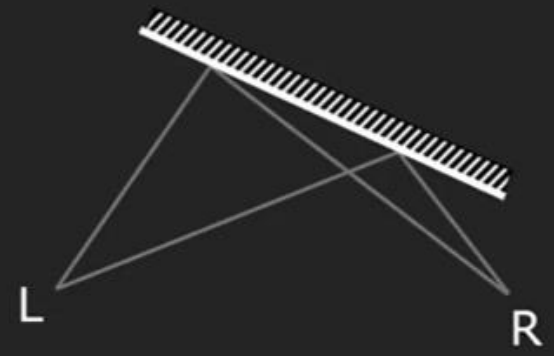
Issues with Stereo Matching

- Surface must have (non-repetitive) texture





- Foreshortening effect makes matching challenging



How Large Should Window Be?



Window size = 5 pixels
(Sensitive to noise)



Window size = 30 pixels
(Poor localization)

Adaptive Window Method Solution: For each point, match using windows of multiple sizes and use the disparity that is a result of the best similarity measure (minimize SSD per pixel).

Window Based Methods: Results



Left Image



Right Image



Ground Truth



SSD (Window size=21)



SSD – Adaptive Window



State of the Art